

No need to submit. Please attempt all these questions yourself and reach out for any difficulties. Most of these questions are from [1]

- Visualize and find the length of the curve $\mathbf{r}(t) = (2t^{3/2} \quad \cos(2t) \quad \sin(2t))$ for $0 \leq t \leq 1$.
- Show that the curve with vector equation

$$\mathbf{r}(t) = (a_1 t^2 + b_1 + c_1 \quad a_2 t^2 + b_2 t + c_2 \quad a_3 t^2 + b_3 t + c_3)$$
 lies in a plane and find an equation of the plane.
- Consider the following three variable functions. For each of these functions, (a) describe the level curves of the two variable functions $f(x, y, 0)$ and (b) describe the level surfaces of $f(x, y, z)$
 - $f(x, y, z) = x + y + z$
 - $f(x, y, z) = x^2 + y^2 + z^2$
 - $f(x, y, z) = x^2 + y^2 + 2z^2$
 - $f(x, y, z) = y^2 + z^2$
- Do the following limits exist? If so, find their values
 - $\lim_{(x,y) \rightarrow (0,0)} y^4 \cos^2 x / (x^4 + y^4)$
 - $\lim_{(x,y) \rightarrow (0,0)} xy^4 / (x^2 + y^8)$
- The temperature (in at a location in the Northern Hemisphere depends on the longitude x , latitude y , and time t , so we can write $T = f(x, y, t)$. Let's measure time in hours from the beginning of January.
 - What are the meanings of the partial derivatives $\partial T / \partial x$, $\partial T / \partial y$ and $\partial T / \partial t$?
 - Honolulu has longitude 158W and latitude 21N. Suppose that at 9:00 AM on January 1 the wind is blowing hot air to the northeast, so the air to the west and south is warm and the air to the north and east is cooler. Would you expect $f_x(158, 21, 9)$, $f_y(158, 21, 9)$ and $f_t(158, 21, 9)$ to be positive or negative? Explain
- Suppose $z = 5x^2y^2$, and (x, y) changes from $(1, 2)$ to $(1.05, 2.1)$. Estimate the change in z by a linear approximation to z . How does this change compare to the actual change in z ?
- Four positive numbers, each less than 50, are rounded to the first decimal place and then multiplied. Use differentials to estimate the error in the computed product that might result from the rounding.
- Find the directional derivative of $f(x, y) = e^x \sin y$ along the direction of the vector $\mathbf{v} = (-6, 8)$.
- The temperature T in a metal ball is inversely proportional to the distance from the center of the ball, which we take to be the origin. The temperature at the point $(1, 2, 2)$ is 120.
 - Find the rate of change of T at $(1, 2, 2)$ in the direction toward the point $(2, 1, 3)$
 - Show that at any point in the ball the direction of greatest increase in temperature is given by a vector that points toward the origin.
- Suppose that the directional derivatives of f are known at a given point in two nonparallel directions given by unit vectors $\mathbf{u}$ and $\mathbf{v}$. Is it possible to find ∇f at this point? If so, how would you do it?
- Find the maxima and minima of $f(x, y) = x^2 y e^{-x^2 - y^2}$ for $x, y \in \mathbb{R}$. Are there any local maxima, minima, or saddle points other than the points you mentioned? Repeat for $f(x, y) = x^2 + 4y^2 - 4xy + 2$ and $f(x, y) = x^4 + y^4 - 4xy$.

References

[1] James Stewart. *Calculus: early transcendentals*. Cengage Learning, 2012.